S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 09

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3)

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Date of Performance:

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❖ CODE :

```
GNU nano 8.7                                                    optimal.c
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12, i, j, k, pos, max, faults = 0, hits = 0;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        if(!found) {
            int farthest = -1;
            pos = -1;

            for(j=0;j<3;j++) {
                int next = -1;

                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {
                        next = k;
                        break;
                    }
                }

                if(next == -1) {
                    pos = j;
                    break;
                }

                if(next > farthest) {
                    farthest = next;
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            faults++;
```

```
^G Help      ^O Write Out   ^F Where Is    ^K Cut       ^T Execute   ^C Locatio
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To L
```

```
GNU nano 8.7

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                found = 1;
                hits++;
                break;
            }
        }

        if(!found) {
            int farthest = -1;
            pos = -1;

            for(j=0;j<3;j++) {
                int next = -1;

                for(k=i+1;k<n;k++) {
                    if(frames[j] == pages[k]) {
                        next = k;
                        break;
                    }
                }

                if(next == -1) {
                    pos = j;
                    break;
                }

                if(next > farthest) {
                    farthest = next;
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            faults++;
        }
    }

    printf("Total Page Hits = %d\n", hits);
    printf("Total Page Faults = %d\n", faults);

    printf("Hit Percentage = %.2f%%\n", (hits/(float)n)*100);
    printf("Fault Percentage = %.2f%%\n", (faults/(float)n)*100);

    return 0;
```

```
^G Help      ^O Write Out   ^F Where Is    ^K Cut
^X Exit      ^R Read File   ^\ Replace     ^U Paste
```

❖ OUTPUT :

```
~

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ nano optimal.c

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ gcc optimal.c -o optimal

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ ./optimal
Total Page Hits = 5
Total Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$
```